CLEMENT ASARE

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LinkedIn Profile | GitHub Portfolio

PROFESSIONAL SUMMARY

Mechanical Engineering graduate with practical experience in industrial maintenance, electro-mechanical systems and technical programming. Combines field experience with hands-on project work in computer vision, AI and microcontroller programming. Seeking to pursue a Master's in Mechatronics or Robotics to deepen my knowledge of control systems, automation and intelligent system design, with a focus on developing practical solutions for agriculture, healthcare and industry.

RESEARCH INTERESTS

- Autonomous Systems & Mobile Robotics
- Embedded Control Systems & Industrial Automation
- Computer Vision & Sensory Perception
- Human-Robot Collaboration

EDUCATION

University of Mines and Technology | BSc (Hons) Mechanical Engineering (CWA: 79.02, 2023)

Final Year Thesis: Design and Energy Analysis of a Micro-Hydro Power System Using Residential Rainwater Storage

- Designed a micro-hydro power generation system (15.4 m elevation head) to produce off-grid electricity.
- Evaluated impulse and reaction turbines and selected a Pelton wheel based on efficiency and feasibility.
- Performed flow calculations, mechanical component sizing and electrical energy analysis achieving power output of 16.33 kW.

Relevant Coursework: Machine Design, Instrumentation and Control, Computer Programming, Mechanics of Machines, Applied Electricity

TECHNICAL SKILLS

Robotics & Controls: ROS2, Embedded Systems, Arduino, Sensor Integration

Mechanical Design: SolidWorks, AutoCAD

Programming Languages: Python, C++, MATLAB, JavaScript, PHP, C#

AI/Data Science: Computer Vision, Machine Learning, NumPy, Pandas, LangGraph, Matplotlib

Databases: MySQL, SQLite, MongoDB

PERSONAL PROJECTS

CLEMVIS — Real-Time Visual Motion Detection and Tracking System |Python, OpenCV, SQLite

- A real-time computer vision and motion detection & tracking system featuring background subtraction, centroid tracking, zone-based monitoring, finite-state machine control and performance evaluation.

Closed-Loop DC Motor Speed Control | Arduino, Python, Numpy

- Developed an ATmega328P-based closed-loop DC motor control system using PID techniques and encoder feedback to maintain precise speed under varying loads.

Conversational AI Personal Assistant | Python, Google Gemini 2.5 flash model, pypdf

- A chatbot that respond to user queries about a person and give responses on behalf of the person.

WORK EXPERIENCE

Applications Engineer (Trainee) | Metso Ghana Ltd | Dec 2024 – Present

- Select and size Metso slurry pumps for mining operations across West Africa, ensuring optimal performance in the plant.
- Troubleshoot, inspect and service slurry pumps, cyclones and control valves to minimize plant operational downtime.
- Install Metso Pump Condition Monitoring system to monitor equipment health through vibration analysis.
- Prepare comprehensive technical reports following field inspections and maintenance activities.

National Service Personnel | Metso Ghana Ltd | Nov 2023 – Oct 2024

- Assisted in maintenance and servicing of MKIII 54-75 gyratory crusher, RTGT filter and slurry pumps.
- Operated a FARO Laser Focus Scanner to scan crusher liners for engineering wear analysis.
- Conducted risk assessments prior to maintenance activities to ensure workplace safety compliance

Student Internship | Tarkwa Municipal Hospital, Tarkwa, Ghana | Apr 2022 – Jun 2022

- Troubleshot and repaired biomedical devices like blood pressure monitors, oxygen concentrators and autoclaves.

LEADERSHIP & VOLUNTEERING ACTIVITIES

Project Member | Enactus Ghana (UMaT Chapter) | 2022-2023

- Worked with other students to design projects for community development including water purification system for a mining town, making water safe for consumption.

Electoral Commissioner | Faculty of Engineering Students Association, UMaT | 2022

- Organized and coordinated the voting process for the election of leaders for the Faculty of Engineering Students Association.

ONLINE CERTIFICATE COURSES

- Agentic AI: Build AI Agents with LangGraph, CrewAI & MCP — Udemy (In Progress)
- Artificial Intelligence Fundamentals — IBM SkillsBuild (2025)
- Fundamentals of Robotics & Industrial Automation — Coursera (2024)
- Complete Arduino Course for Beginners — Udemy (2024)